Library Reservation

Amith Kumar Das Orko
Rachael Eapen
Lilly Jackson

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Our Project

Study Room Reservation:

- Issues
  - Timing
  - Location
  - Size and Types of Rooms
  - Layout
- Solution
  - Clean & Simple UI
  - Filtering

005 - Media Prep & Digital Collaboration

Type: Miscellaneous

Location: Atrium

5 Seats

Select

133 - Presentation Practice

Type: Group Study

Location: 1st Floor

4 Seats

Select

135 - Knowledge Market Consultation

Type: Miscellaneous

Location: 1st Floor

6 Seats

Select

204 - Study Room

Type: Group Study

Location: 2nd Floor

6 Seats

Select

216 - Seminar Room

Type: Seminar

Location: 2nd Floor

12 Seats

Select

Individual Study Pod 1

Type: Group Study

Location: 2nd Floor

1 Seats

Select


Podcasting
Booth





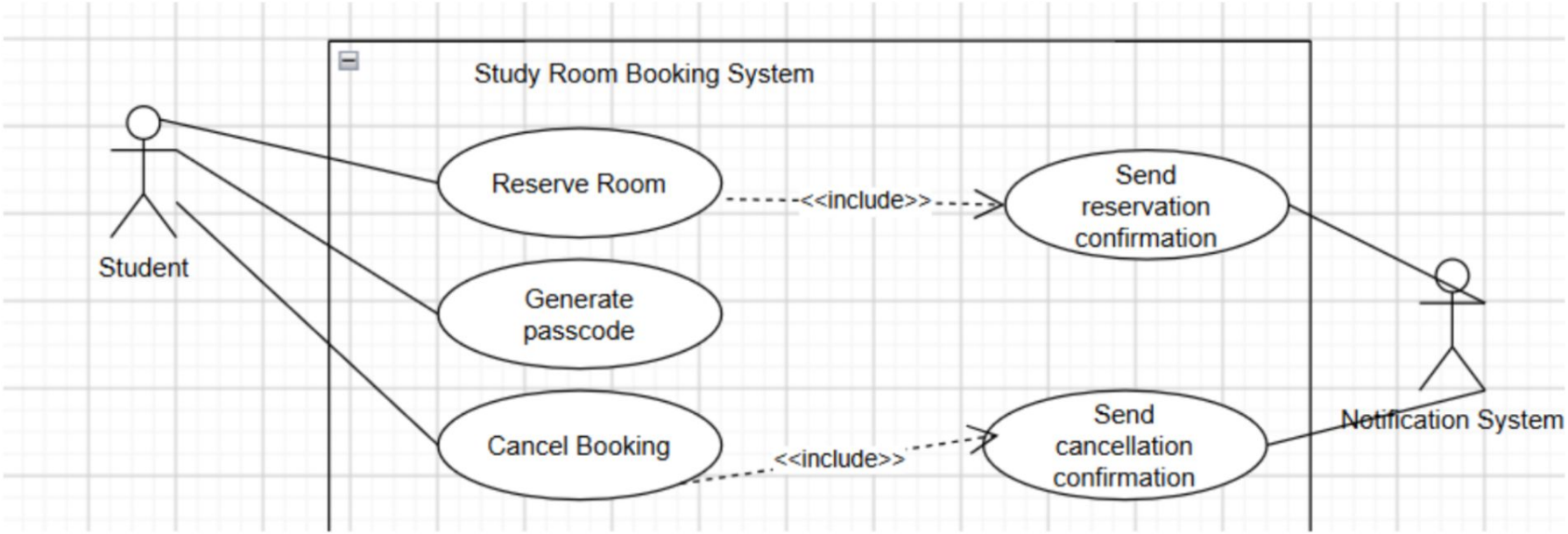
Project Functionalities

Key Features:

1. Availability
2. Reservation
3. Check-in
4. Cancellation

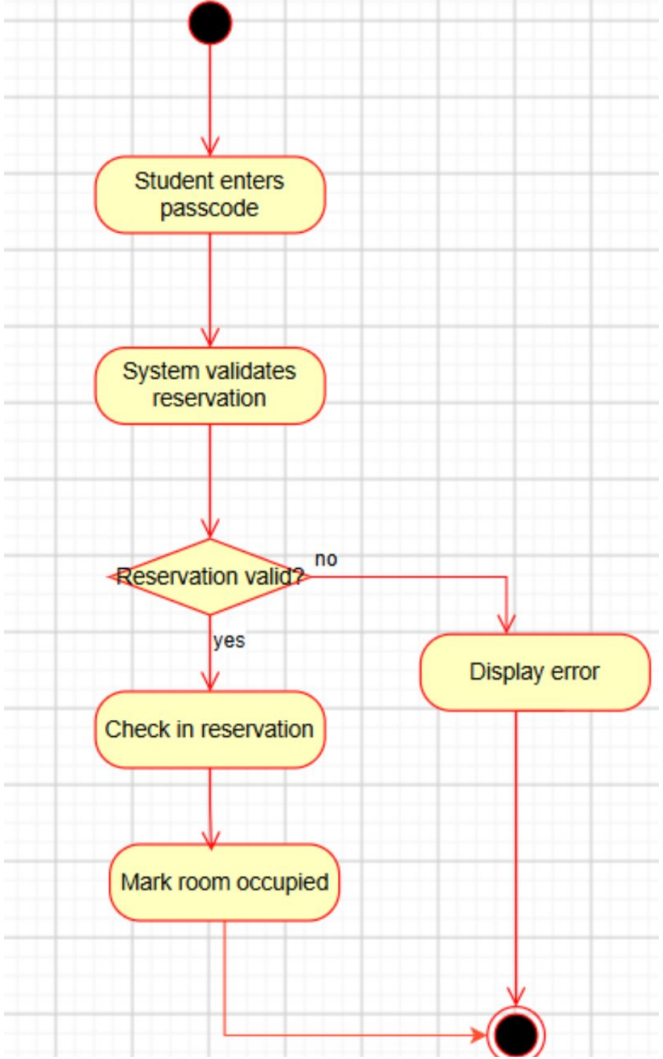
Use Case Diagram

Use case diagram for Study Room Booking System

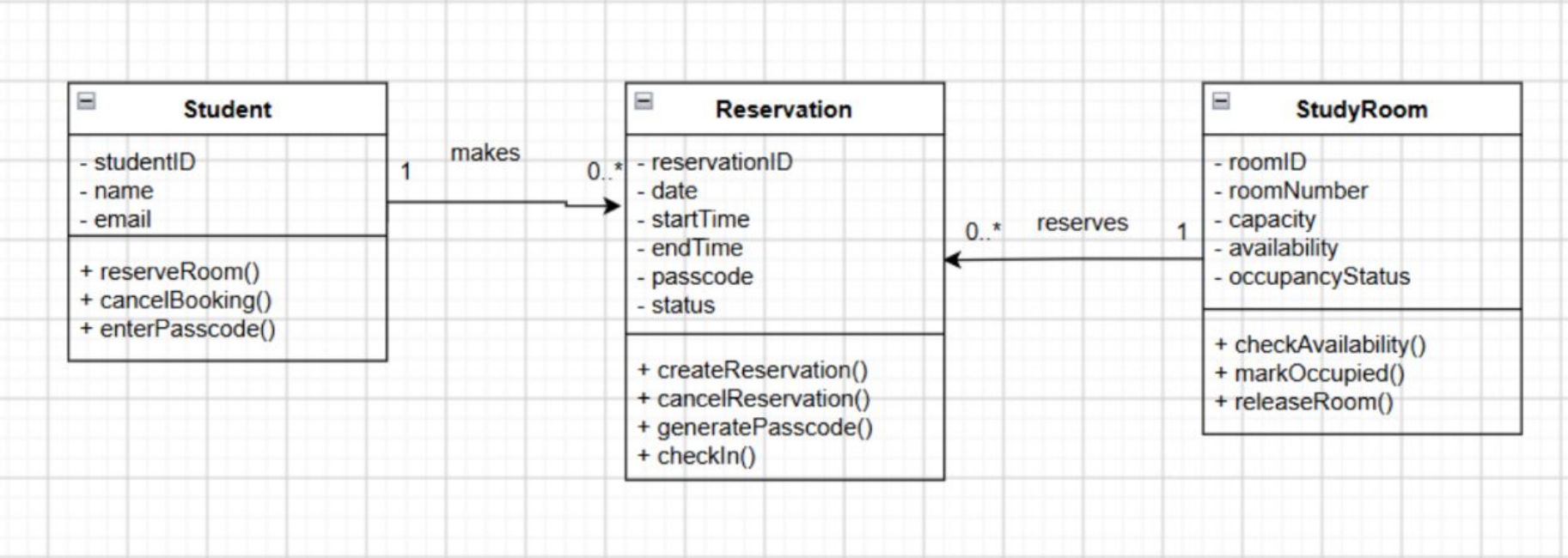


Activity Diagram

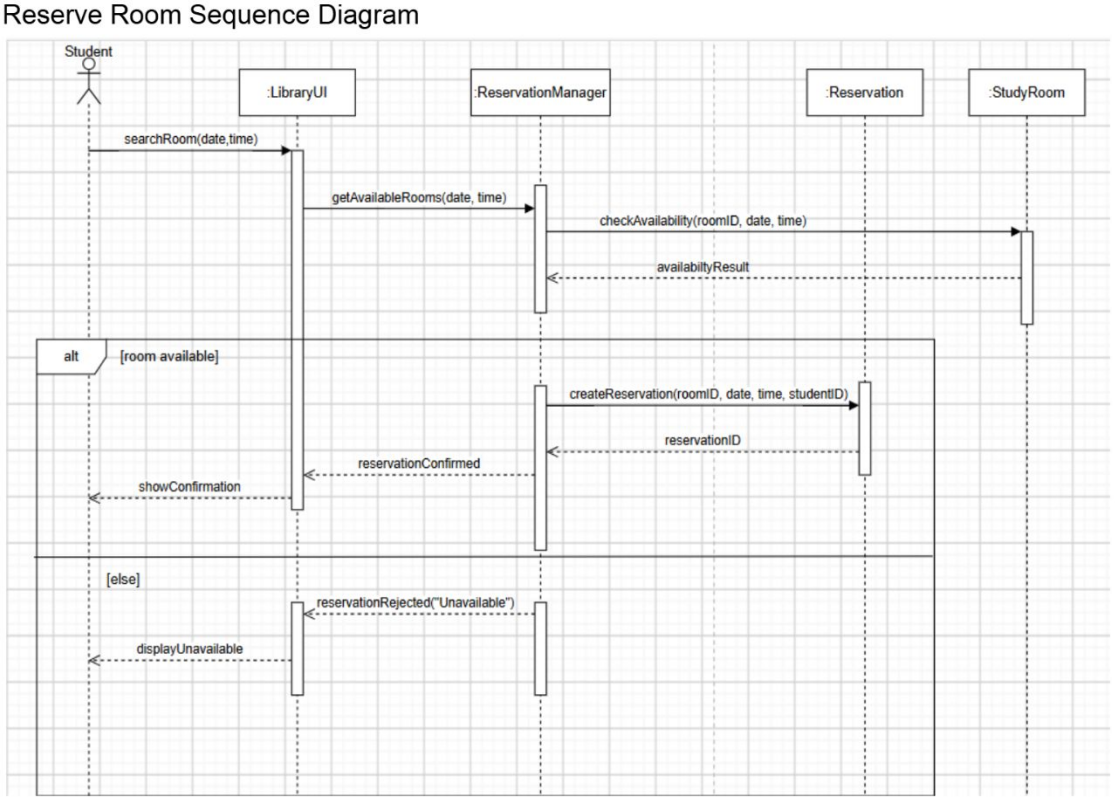
Room check in



Class Diagram



Sequence Diagram



Technology and Team Responsibilities

- Python: Reservation and cancellation logic
- PostgreSQL: Database management and storage
- Streamlit: Front end Development

Team Responsibilities :

1. Orko: Python backend development and database connectivity
2. Rachael: PostgreSQL tables and database queries
3. Lilly: Frontend development , and connecting the interface to backend

Shared: Requirements analysis, UML diagrams, project documentation.

Current Version



Next Steps

- **Double-Booking Prevention:** Test same room reservation
- **Availability Checking:** Display only available rooms
- **Testing:** Functional and integration testing
- **Check-In Feature:** For arriving students
- **Reservation Validation:** Check room capacity etc.
- **Front-end:** Integrate user interface with the Python backend.

Thank You!